AML in Adults

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70.1 AML in CR1

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70.1.1 Definition, Subtypes

AML is a malignancy of hematopoietic immature precursors (myeloblasts) that accumulate in the BM at the expense of their normal counterparts. AML is increasingly being recognized as a heterogenous malignancy based on distinct disease biology and underlying cytogenetic and molecular profiles. Although a blast count of ≥20% in the BM or PB is still considered to diagnose AML, the 2022 World Health Organization (WHO) classification and International Consensus Classification (ICC) considered a lower threshold to allow for disease-defining genetic abnormalities and to acknowledge the biologic continuum between MDS and AML (Khoury et al. 2022; Arber et al. 2022). A less strict definition of blast cutoff in the updated WHO classification and ICC emphasizes a comprehensive assessment of morphology, cytogenetic, and molecular genetic analyses to classify MDS and/or AML.

The ELN recommendations have also been updated in 2022, which stratified patients into three risk groups, including favorable risk, intermediate risk, and adverse risk, based on pretreatment cytogenetic abnormalities and gene mutations (Dohner et al. 2022). The ELN 2022 further recognizes the underlying disease biology in its risk classification by adding a secondary type, MDS-associated mutations to the adverse-risk category (Dohner et al. 2022).

70.1.2 Clinical Presentation

The median age at diagnosis is approximately 70 years, and the annual age-standardized incidence rate varies between 3 and 4 cases per 100,000. Patients with AML typically present with symptoms such as fatigue, bruising, or infections, whereas lymphadenopathy and/or hepatosplenomegaly may be found by physical examination. Analysis of blood work often

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reveals thrombocytopenia, anemia, and/or neutropenia. In some patients, a serious bleeding diathesis can occur, particularly in the early phase of treatment, because the leukemic blasts are able to activate the coagulation cascade and cause hyperfibrinolysis.

70.1.3 First-Line Treatment

Achieving and maintaining a first CR is crucial in AML patients, but treatment may largely fail because of relapse from CR rather than primary induction failure or treatment-related mortality (TRM). The standard intensive AML induction treatment has consisted of 7–10 days of the antimetabolite cytosine arabinoside (Ara-C) and 3 days of an anthracycline (i.e., daunorubicin or idarubicin) since the 1980s. CR rates with standard induction estimate between 70 and 90%. Favorable-risk patients have relatively good outcomes with overall survival rates of approximately 60–70%, whereas outcomes for patients with intermediate-risk and particularly adverse-risk AML remain unsatisfactory.

Targeted therapies have been introduced in AML, of which the kinase inhibitor midostaurin added to intensive induction and consolidation chemotherapy has been approved for first-line treatment of AML patients with mutated *FLT3* based on the survival benefit observed in the RATIFY trial (Stone et al. 2017). Other promising targeted therapies against *FLT3* (e.g., gilteritinib and quizartinib) or *IDH1/2* mutations (e.g., ivosidenib and enasidenib) are currently being investigated in first-line intensive induction treatment of which results are awaited. Alternative induction strategies might consist of gemtuzumab-ozogamicin, a humanized anti-CD33 antibody–drug conjugate or CPX-351, a liposomal formulation of cytarabine and daunorubicin, for specific subgroups of patients. CPX-351 has been shown to improve overall outcome compared with standard intensive induction chemotherapy in patients aged 60–75 years with therapy-related or secondary-type AML, including MDS-related cytogenetic abnormalities (Lancet et al. 2018).

Treatment options have increased considerably for AML patients deemed not eligible for intensive treatment to induce the first CR. Hypomethylating agents, including azacytidine and decitabine, may be a treatment option in mono or combination therapy. Venetoclax, a BCL2 inhibitor, combined with azacitidine increased response rates, which were rapid and durable, and improved survival and quality of life compared with azacitidine alone (DiNardo et al. 2020). As a result, azacitidine and venetoclax treatment has been established as the new standard of care for first-line treatment of older or unfit AML patients. Alternatively, for patients with mutated *IDH1*, ivosidenib combined with azacitidine was associated with higher response rates and superior survival compared with azacitidine alone in a randomized phase III trial (Montesinos et al. 2022). These targeted therapies have different toxicity profiles compared with conventional chemotherapy and need specific monitoring of drug–drug interactions and adverse events such as cardiac QTc assessment, differentiation syndrome, and prolonged cytopenias.

Once a remission is being obtained, post-remission treatment decisions are directed by the ability of pretreatment features such as those incorporated in the ELN risk classification to predict outcome. However, probably more prognostically important than the pretreatment features is response to treatment and especially the presence, in hematological remission, of measurable residual disease (MRD) as assessed by flow cytometry or targeted quantitative PCR for specific markers.

70.1.4 HCT and AML Risk Categories

70.1.4.1 AML Risk Categories

Previously, conventional cytogenetics and mutations in *NPM1*, *FLT3*-ITD, *CEBPA*, *RUNX1*, *ASXL1*, and *TP53* genes were included in the ELN 2017 risk classification of AML patients (Dohner et al. 2017). The current ELN 2022 risk classification has added mutations in *BCOR*,

EZH2, *SF3B1*, *SRSF2*, *STAG2*, *U2AF1*, and/or *ZRSR2*, which are being recognized as MDS-related mutations with an adverse outcome (Table 70.1) (Dohner et al. 2022). Other important changes include (1) the *FLT3*-ITD allelic ratio not being considered in the classification; (2) only in-frame mutations in the b-zip region of *CEBPA* are now considered as favorable risk, irrespective of being mono- or bi-allelic; (3) *NPM1* mutations are not classified as favorable if adverse-risk cytogenetics are present; (4) recurring cytogenetic abnormalities including t(3q26;v) or t(8;16)(p11.2;13.3) are being included in the adverse-risk group; (5) hyperdiploid karyotypes are not considered complex; and (6) only *TP53* mutations at a variant allele fraction of≥10% are considered adverse risk.

Similar to the previous risk classification, the ELN 2022 AML risk classification is advocated to be used for risk-stratifying AML and to a risk-adapted treatment approach for patients with AML. Such a risk-adapted treatment approach for patients with AML depends not only on the risk of relapse of the underlying AML but also on the risk of TRM associated with the applied post-remission treatment. The application of MRD, detected by either multiparametric flow cytometry or quantitative PCR for specific molecular markers may further improve AML risk classifications. MRD may be detected at time points

Table 70.1 Risk-adapted post-remission treatment for patients with AML in the first CR[a]

AML risk classification[b]	MRD status[c]	Preferred post-remission treatment
Favorable		
t(8;21)(q22;q22.1); *RUNX1-RUNX1T1*	Negative	Chemotherapy/auto-HCT
Inv(16)(p13.1q22) or t(16;16)(p13.1;q22); *CBFB-MYH11*		
Mutated *NPM1* without *FLT3*-ITD (without adverse-risk cytogenetic abnormalities)	Positive	Allo-HCT[b] (unless excessive TRM can be predicted)
bZIP in-frame mutated *CEBPA*		
Intermediate		
Mutated *NPM1* with *FLT3*-ITD (without adverse-risk cytogenetic abnormalities)	Negative	Consider allo-HCT[b] (if acceptable risk of TRM; alternative, chemo/auto-HCT)
Wild-type *NPM1* without *FLT3*-ITD (without adverse-risk genetic lesions)		
t(9;11)(p21.3;q23.3); *MLLT3-KMT2A*	Positive	Allo-HCT[e]
Cytogenetic and/or molecular abnormalities not classified as favorable or adverse		
Adverse		
t(6;9)(p23;q34.1); *DEK-NUP214*	Negative	Allo-HCT[e]
t(v;11q23.3); *KMT2A* rearranged		
t(9;22)(q34.1;q11.2); *BCR-ABL1* t(8;16)(p11.2;p13.3);*KAT6A-CREBBP*	Positive	Allo-HCT[e]
Inv(3)(q21.3q26.2) or t(3;3)(q21.3;q26.2); *GATA2, MECOM(EVI1)* t(3q26.2;v)/*MECOM*(*EVI1*)-rearranged		
−5 or del(5q); −7; −17/abn(17p)		
Complex karyotype (excluding hyperdiploid), monosomal karyotype		
Mutated *ASXL1*, *BCOR*, *EZH2*, *RUNX1*, *SF3B1*, *SRSF2*, *STAG2*, *U2AF1*, and/or *ZRSR2*		
Mutated *TP53* with VAF ≥10%		

[a]Adapted from (Cornelissen et al. 2012b), Table 4
[b]Allo-HCT using HLA-identical sibling or 10/10 MUD donors
[c]Adapted from (Dohner et al. 2022), Table 6
[d]Detected with multiparametric flow cytometry or with for qPCR specific markers
[e]Allo-HCT using HLA-identical sibling, MUD, MMUD umbilical cord blood, or haploidentical donors

early after induction treatment to assess the remission status of the AML but also after post-remission treatment to detect imminent relapse. Consequently, MRD negativity has been introduced as an end point in patients with a hematological CR (Dohner et al. 2017).

70.1.4.2 Transplantion Risk Categories

The risk-adapted approach of patients with AML in the first CR should also include the assessment of TRM for each individual patient. TRM may be attributed to GVHD, infectious complications, organ toxicity, and other causes (Penack et al. 2020). A number of parameters may relate to allo-HCT-related mortality, including the procedure (e.g., conditioning regimen and application of TCD), donor characteristics (e.g., HLA-matching), and recipient features (e.g., age and comorbidity). The risk of mortality may be quantified by composite risk scores, which have been established to predict TRM and overall outcome.

Two generally approved transplant risks were developed and validated, including the EBMT risk score (Gratwohl et al. 1998) and the hematopoietic cell transplantation-comorbidity index (HCT-CI) (Sorror et al. 2005). The EBMT risk score is based on patient and transplantation characteristics, which was developed in CML patients and subsequently validated in other patient groups including AML (Gratwohl et al. 2009). The HCT-CI consists of 17 comorbidities that contribute to a cumulative score, which has been extensively validated and continuously being refined, including age, disease status, or biomarkers (Sorror et al. 2007, 2014). A machine-based learning model was developed by the EBMT-acute leukemia working party (ALWP) based on 10 variables, which was highly predictive of mortality at 100 days and 2 years (Shouval et al. 2015). Other groups have also developed predictive models for TRM modifying the weights of the EBMT risk score and the HCT-CI (Barba et al. 2010), whereas others combined transplant-related parameters and patient characteristics or used biomarkers (Parimon et al. 2006; Barba et al. 2014; Luft et al. 2017).

Developments in allo-HCT care have improved outcomes over the last decades, attributable to for example the introduction of reduced intensity conditioning, improved supportive care and infectious prophylaxis, and better GVHD prophylaxis or GVHD treatment. Consequently, TRM has been strongly reduced and the use of allo-HCT as post-remission treatment for older or less fit patients with comorbidities has increased (Snowden et al. 2022). Several groups have reported less predictive power of the established TRM risk models in the current era of allo-HCT. The EBMT-ALWP has developed updated scoring systems applicable to the setting of RIC in older patients with AML and PTCY as GVHD prophylaxis which both had increased predictive power (Versluis et al. 2015; Hermans et al. 2023). Similar to updating AML risk classifications, the prediction of TRM also needs continued refinement and reassessment in specific patient groups and novel treatments.

70.1.5 HCT in First-Line AML Treatment: A Risk-Adapted Approach

AML risk classifications are being used for tailoring patients' optimal post-remission treatment, which include allo-HCT, high-dose chemotherapy followed by auto-HCT, and continued chemotherapy. Allo-HCT is the most optimal post-remission treatment for the prevention of relapse due to a potent GVL effect, which has been demonstrated to be exerted irrespective of underlying AML cytogenetic subcategories and MRD status (Cornelissen et al. 2012a; Versluis et al. 2017a). However, absolute estimates of relapse incidence differ and reflect molecular or cytogenetic differences resulting in resistance of the AML. Although the GVL effect of allo-HCT is unequivocally present in patients with AML in the first CR, concurrent TRM compromises overall outcome, especially in AML patients with a relatively low incidence of relapse. Thus, a risk-adapted approach of post-remission treatment for patients with AML in the first CR should include an assessment of the TRM risk in addition to leu-

kemia characteristics and MRD (Cornelissen et al. 2012b; Cornelissen and Blaise 2016). Table 70.1 summarizes a risk-adapted approach based on the ELN 2022 AML risk classification, MRD status, and the risk for TRM. The risk for TRM should be preferably assessed with dedicated scores for specific subgroups of patients.

Allo-HCT is generally not indicated in patients with a favorable AML risk profile; for those patients auto-HCT or continued chemotherapy may be preferred (Cornelissen et al. 2012b; Cornelissen and Blaise 2016; Dohner et al. 2022). However, favorable-risk patients with MRD are considered high risk for relapse and preferably receive an allo-HCT in the first CR, unless excessive TRM is predicted even with RIC.

Results of allo-HCT compared with non-allo-HCT post-remission therapies have yielded contradicting results in intermediate-risk patients, especially taking molecular markers into account (Koreth et al. 2009; Stelljes et al. 2014; Versluis et al. 2017b). A risk-adapted AML trial which allocated patients with MRD negative, intermediate-risk AML to auto-HCT, whereas intermediate-risk patients with MRD positivity received allo-HCT, showed similar outcomes for auto-HCT and allo-HCT (Venditti et al. 2019). Consequently, assessment of MRD status is strongly advocated for post-remission treatment decisions in patients with an intermediate-risk AML. Allo-HCT may be applied in patients with intermediate-risk AML with MRD after induction chemotherapy, except for patients with a high risk for TRM. Allo-HCT might also be considered for patients with intermediate risk, MRD negative AML, but auto-HCT and chemotherapy are alternative treatment options, particularly when the predicted risk for TRM is high.

Adverse-risk patients with MRD should be transplanted with an allograft as soon as a hematological CR is obtained. Adverse-risk patients without MRD still have a significant risk of relapse and are also candidates for allo-HCT, although patients with a very high risk for TRM may be considered for non-allo-HCT approaches.

70.2 Allo-HCT in Advanced AML

Charles Craddock

70.2.1 Introduction

Allo-HCT plays an increasingly important role in the management of AML in adults (Dohner et al. 2022). The advent of RIC regimens coupled with increased donor availability has dramatically increased the number of patients in whom allo-HCT can be contemplated. At the same molecular characterization at diagnosis coupled with MRD, quantitation after induction chemotherapy has considerably improved our ability to predict relapse risk in patients treated with intensive chemotherapy alone refining our ability to identify patients with the potential to benefit from allogeneic transplantation (Loke et al. 2020). Consequently, allo-HCT is now a pivotally important personalized component of the treatment algorithm in adults with AML in CR1. At the same time, there is an emerging recognition of the curative potential of allografting in patients with advanced-phase AML, particularly patients with primary refractory (PREF) AML (Dohner et al. 2022). It is therefore extremely important that all fit adults with newly diagnosed AML are tissue typed at presentation and, unless there are compelling reasons to believe allo-HCT will not be included in the patients' treatment algorithm, a donor search is commenced in a timely manner. Encouraging results have been achieved using both matched sibling and volunteer unrelated donors and more recently transplantation using haploidentical donors (in conjunction with post-transplant cyclophosphamide GVHD prophylaxis) and cord blood units with a high nucleated cell dose. To date, there is no compelling data concerning whether there is an optimal stem cell source and prospective studies are required to address this important point. In both patients transplanted in CR1 and those with advanced phase disease, there has been a substantial reduction in transplant-related mortality (TRM) over the decades, but the risk of disease relapse post-

transplant remains stubbornly high and now represents the major cause of treatment failure in patients allografted for AML. There remains, therefore, an urgent requirement to develop novel strategies with the potential to reduce the risk of disease recurrence in all patients allografted for AML.

70.2.2 The Role of Allo-HCT in the Management of Primary Refractory AML

Ten to forty percent of adults with newly diagnosed AML have PREF AML defined as a failure to achieve a morphological CR after two courses of intensive induction chemotherapy, including at least one cycle of intermediate dose cytarabine (Ferguson et al. 2016). Factors determining refractoriness to induction chemotherapy include adverse disease biology and patient age. Patients with PREF AML have genuinely chemo-refractory disease with long-term survival rates <10% if treated with further chemotherapy, and allo-HCT represents the only treatment modality with the potential to deliver long-term disease-free survival. Evidence that allo-HCT can deliver long-term survival in a significant proportion of patients with PREF AML has been accumulating over the last decade and represents an important advance in the management of this sizeable patient population for whom no other effective therapy exists (Craddock et al. 2011; Todisco et al. 2017; Brissot et al. 2017; Boyiadzis et al. 2023). Nonetheless, outcomes in patients allografted for PREF AML remain unsatisfactory, and both TRM and disease relapse continue to represent significant barriers to long-term survival. There is also a lack of clarity concerning which patients with PREF AML are the most likely to benefit from transplantation. Outcome is superior in patients who proceed swiftly to transplant after no more than two courses of intensive chemotherapy, and relapse appears to be lower in those with a lower burden of disease at the time of transplantation. It is therefore important to consider transplant as a potentially curative treatment modality in fit patients with PREF AML, providing a suitable donor can be identified rapidly and the predicted TRM is acceptable. However, further studies both on the impact of disease biology and evidence of chemo-responsiveness on outcome after allo-HCT are required if we are to be able to identify with greater clarity both which patients with PREF AML are likely to benefit from allo-HCT and equally for whom this further intensification in treatment is unlikely to deliver long-term survival. At present, accumulating data identifies very poor outcomes for patients with high-risk molecular features such as the presence of a *TP53* mutation or inv(3)(q21.3q26.2) and on the basis of current knowledge it is probably reasonable to reserve allogeneic transplantation in this patient group for patients who have achieved a morphological CR (Loke et al. 2022; Daver et al. 2022; Badar et al. 2023).

The optimal conditioning regimen in patients with PREF AML remains a matter of conjecture, but it is recommended that a myeloablative regimen is utilized in patients under the age of 50 years unless there are compelling reasons to the contrary. In older adults, there is no evidence that any particular RIC regimen is to be favored. Although encouraging results using a FLAMSA sequential regimen were reported in patients with PREF AML, it is relevant to note that a recent randomized trial failed to demonstrate any benefit of such a regimen compared with the standard RIC regimen FB2 in adults with AML in CR1 or CR2, although insufficient patients with PREF AML were randomized to address whether this regimen has particular activity in this clinical setting (Craddock et al. 2021). As with all patients allografted for high-risk AML, it is important to assess response to transplant by performing MRD studies on a bone marrow aspirate 30–45 days posttransplant, since this may guide the implementation of an early immunosuppression taper or use of prophylactic/pre-emptive donor lymphocyte infusion (DLI) (Loke et al. 2023).

70.3 Practical Issues for Allo-HCT in AML

Arnon Nagler

AML is the most frequent indication in Europe for allogeneic HCT. As per the latest publication by Passweg et al., 7123 allo-HCT has been performed in Europe in 2021 for AML (4266 for de novo AML in the first CR, 1775 for de novo AML not in CR, and 1082 for AML-therapy-related or myelodysplasia-related changes) accounting for 38% of the allogeneic transplant activity in Europe (Passweg et al. 2021). In sharp contrast to the lymphatic malignancies, the allogeneic activity in AML has not been decreased as CAR-T cell treatment for AML is still in its infancy (Passweg et al. 2021). As in many of the AML patients with an indication for allo-HCT, transplant is currently the only curative procedure and in those achieving CR after first induction results are better. It is therefore recommended to perform HLA typing of the patient and the potential family donors upfront once the AML is diagnosed and to start a search (Carreras et al. 2019). In recent years, continuous novel developments in the field of allo-HCT and mainly the reduced intensity/reduced toxicity conditioning and haploidentical donor allo-HCT with post-transplant cyclophosphamide (PTCY) have significantly reduced transplant-related toxicities and mortality combined with the improved supportive care (Slavin et al. 1998; Kasamon et al. 2015; Shimoni et al. 2005). Consequently, the median age of the AML patients that are undergoing transplantations is increasing with many of the transplanted patients being over 60 years or even 65 years of age. Many of these patients are receiving the allo-HCT following upfront induction with low-intensity Venetoclax-based protocols which further reduced toxicities (Dohner et al. 2022; Keith et al. 2022). Achieving measurable residual disease (MRD) negativity, pre-allo-HCT correlates with outcomes and becomes a major goal in the treatment paradigm for acute leukemias and other hematological malignancies as well (Keith et al. 2022; Abhishek et al. 2021; Bazinet et al. 2023). Most of the allo-HCTs that are being performed nowadays are from unrelated including mismatched unrelated and haploidentical donors and with mobilized PBSC (Nagler and Mohty 2022; Arslan and Al Malki 2022; Anasetti et al. 2012; Raghunandan 2022). Patients with a high risk of relapse post-allo-HCT including those with high-risk AML, secondary poor prognostic mutations, or secondary AML should be considered for maintenance or preemptive therapy post-allo-HCT in an attempt to reduce the relapse rates (Eshrak et al. 2023). Finally, PTCY-based regimens are becoming the preferred mode of GVHD prophylaxis including for patients with AML undergoing HLA-matched related, unrelated, and haploidentical allo-HCT, and it may change other aspects of allo-HCT including the intensity of the conditioning as well as the risk of relapse post-transplantation (Bolanos-Meade et al. 2023).

70.4 Acute Promyelocytic Leukemia

Miguel Á. Sanz

70.4.1 Concept and Incidence

APL is a subtype of AML with peculiar clinical and morphological characteristics, which presents a specific genetic alteration, the t (15; 17), with its corresponding molecular counterpart, the rearrangement *PML-RARA*, which confer a particular sensitivity to all-*trans* retinoic acid (ATRA) and arsenic trioxide (ATO). It also highlights the presence of a hemorrhagic diathesis associated with a peculiar coagulopathy, which causes a high incidence of hemorrhagic complications at presentation and early during the induction treatment.

APL accounts for 5–15% of AML, with a median age of around 40 years and similar distribution by sex. About 10% occur after the use of cytotoxic drugs (especially topoisomerase II inhibitors) or radiation.

70.4.2 Diagnosis

70.4.2.1 Morphology, Immunophenotyping, and Other Features

M3 typical (hypergranular) 75–80%	M3 variant (microgranular) 20–25%
Morphology	
• Cytoplasm with dense granulation. Frequent Auer rods • Reniform or bilobed nucleus	• Cytoplasm with fine granulation or hypogranular. Less frequent Auer rods • Reniform nucleus, bi- or multilobed
Immunophenotyping	
HLA-DR^{-}/CD34^{-}/CD33^{+a}/CD13^{+b}/CD15$^{-/+}$	HLA-DR$^{\pm}$/CD34$^{\pm}$/CD33^{+a}/CD9^{+}/CD2$^{\pm}$/CD13^{+b}/CD56 $^{\pm}$
Other associated features	
• Most frequently, low WBC counts • Less frequently, BCR3 isoform	• Most frequently, high WBC counts • Most frequently, BCR3 isoform

[a]Intense and homogeneous expression
[b]Heterogeneous expression

70.4.2.2 Genetic Diagnosis

Conventional cytogenetics t(15;17)(q22;q21)
Pros – Very specific – Detects additional anomalies in 30% (+8 the most frequent) *Cons* – Low sensitivity (80%) – Inadequate, bad metaphases or normal karyotype (false-negative) in 20%
FISH *PML-RARA*
Pros – Very specific and rapid *Cons* – Not very sensitive and does not provide information about the isoform
RT-PCR
Pros – Very specific, rapid, and sensitive – Identifies the isoform, which allows MRD monitoring *Cons* – Occasional artifacts and contaminations
Immunostaining with anti-PML antibody (PG-M3)
Pros – Very specific, rapid, and cheap – Characteristic microspeckled pattern by indirect immunofluorescence *Cons* – Does not provide information about the isoform

70.4.2.3 Other Rearrangements of the RARA Gene on Chromosome 17

Chromosomal abnormality	*RARA* rearrangement
• t(11;17) (q23;q21) • t(17;17) (q21;q21) • t(11;17) (q23;q21) • t(5;17) (q35;q21) • t(11;17) (q13;q21) • t(17;17)(q21;q24) • t(X;17)(p11;q21 • t(4;17) (q12;q21) • t(2;17) (q32;q21) • t(3;17) (q26;q21) • t(7;17) (q11;q21) • t(1;17) (q42;q21) • t(3;17)(q26;q21) • der(17) • ins(15;17)(q22;q21;q25) • t(1;17)(q3;q21)	• *ZBTB16 (PLZF)/RARA* (ATRA-resistant) • *STAT5b/RARA* (ATRA-resistant) • *KMT2a/RARA* (ATRA sensitivity unknown) • *NPM1/RARA* (ATRA sensitivity unknown) • *NUMA1/RARA* (ATRA sensitivity unknown) • *PRKAR1A/RARA* (ATRA sensitive) • *BCOR/RARA* (ATRA sensitive in two cases) • *FIP1L1/RARA* (ATRA sensitivity unknown) • *NABP1 (OBFC2A)/RARA* (ATRA sensitive in one case) • *TBLR1/RARA* (insensitive to ATRA) • *GTF2l/RARA (*ATRA sensitive) • *IRF2BP2/RARA* (ATRA sensitive) • *FNDC3B/RARA* (ATRA sensitive) • *STAT3/RARA* (ATRA sensitivity unknown) • *PML/ADAMTS17/RARA* (ATRA sensitivity unknown) • *THRAP3/RARA* (ATRA-resistant)

70.4.3 First-Line Treatment

The ELN recommendations in 2009 acknowledged the promising outcomes reported in several non-randomized studies using ATRA plus ATO, with or without minimal use of chemotherapy, but the standard of care remained the combination of ATRA plus anthracycline-based chemotherapy (Sanz et al. 2009). However, recent findings have led to modify this recommendation in the updated ELN guidelines (Sanz et al. 2019).

The long-term results of a non-randomized study (Abaza et al. 2017) and two randomized clinical trials (Lo-Coco et al. 2013; Burnett et al. 2015), comparing the efficacy and safety of ATRA plus ATO versus the standard ATRA plus chemotherapy approach, strongly support the former combination as the new standard of care for low-to-intermediate-risk APL patients with WBC counts lower than 10×10^9/L at presentation. Nevertheless, in countries where chemotherapy is more affordable than ATO, the classical combination of ATRA and chemotherapy remains an acceptable option. For high-risk patients, however, there are two valid options, either ATRA plus chemotherapy or ATRA plus ATO with a certain amount of cytoreductive chemotherapy, at least during the induction phase.

HCT is never indicated in patients in CR1, except for the small fraction of patients with persistent RQ-PCR positivity of *PML-RARA* after consolidation (<1%), given the poor prognosis associated with this subset. HCT is also indicated in APL patients who relapse and achieve second or subsequent CR.

70.4.4 Salvage Therapy

Apart from patients with MRD positivity at the end of consolidation (molecular persistence), there is a consensus that patients experiencing molecular or hematological relapse later on require immediate additional treatment, including HCT. The goal of salvage treatment is to achieve molecular remission as a bridge to HCT. In cases where ATRA plus chemotherapy was the frontline therapy, salvage treatment with ATRA plus ATO is recommended. Conversely, if frontline therapy involves ATRA plus ATO, ATRA plus chemotherapy is the preferred option.

The use of gemtuzumab ozogamicin may also be considered in both situations, but always as a bridge to HCT. Various studies (Yanada et al. 2013; Holter Chakrabarty et al. 2014; Lengfelder et al. 2015) suggest that auto-HCT should be considered the first choice for eligible patients achieving a second molecular remission. Patients who are unsuitable for HCT or have a very prolonged CR1 can be managed with some maintenance therapy, which would be chosen considering previous treatments and clinical condition.

Allo-HCT should be reserved for patients with a high risk of relapse and low risk of TRM, but also as a second option for patients who relapse after an auto-HCT.

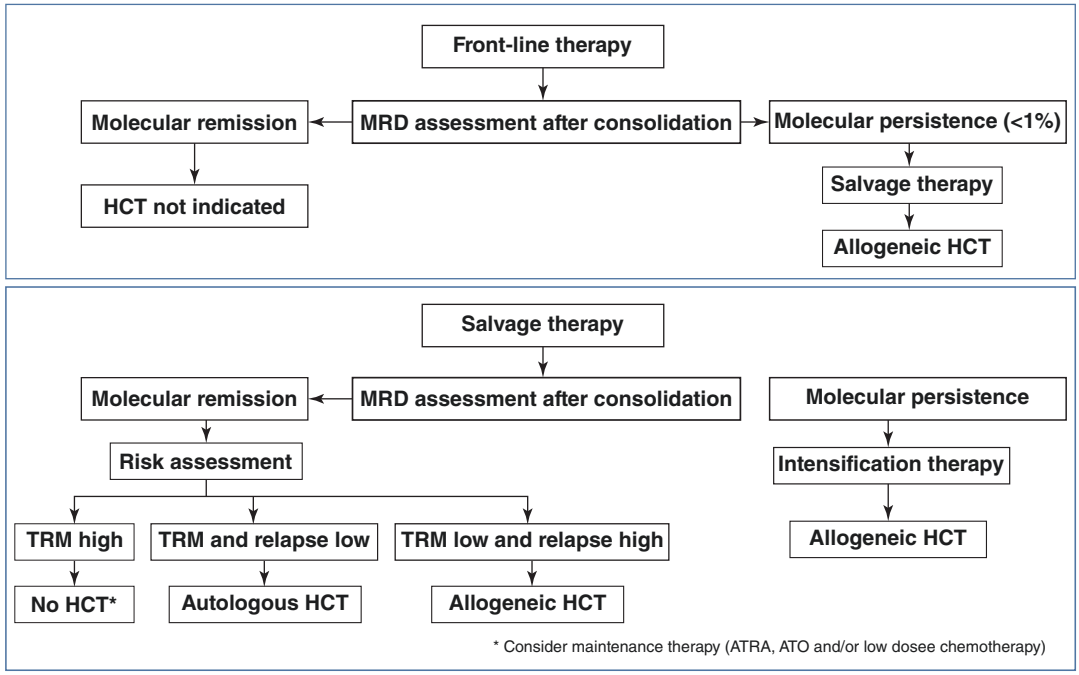
70.4.5 Indications of HCT

HCT is never indicated in patients in CR1, with the exception of those who fail to achieve molecular remission at the end of consolidation (<1%). For a comprehensive overview of HCT indications and other recommendations for patients who require HCT, please refer to Table 70.2 and the algorithm depicted in the following figure.

70.4.6 Main Series Reported on HCT in APL

There are no randomized trials to evaluate the efficacy and safety of the different modalities of HCT in refractory/relapsed APL. The data come mostly from retrospective studies comparing historical cohorts from registries (Table 70.3). Registry studies of both the EBMT (Sanz et al. 2021) and the CIBMTR (Holter Chakrabarty et al. 2014) showed superiority in overall survival for autologous compared to allogeneic HCT, although there were no significant differences in the relapse rate. Other non-comparative JALSG studies have also recently reported the remarkable efficacy of autologous HCT after ATO salvage therapy (Yanada et al. 2013, 2017).

Table 70.2 Indications of HCT in patients with APL



	Auto-HCT	Allo-HCT
Not indicated	CR1 in molecular remission	CR1 in molecular remission
Indicated	≥CR2, but in molecular remission	– ≥CR2 with *PML-RARA* (+) after salvage therapy – ≥CR2 if an auto-HCT has failed previously – ≥CR2 in patients with a high risk of relapse and low risk of TRM
Salvage therapy as a bridge to HCT	Attempt to achieve molecular remission with ATRA plus ATO in patient who relapsed after ATRA plus chemotherapy as front-line therapy, whereas ATRA plus chemotherapy is the option when patients relapse after ATRA plus ATO	Attempt to achieve molecular remission with ATRA plus ATO in patient who relapsed after ATRA plus chemotherapy as front-line therapy, whereas ATRA plus chemotherapy is the option when patients relapse after ATRA plus ATO
Conditioning regimen	Either for use in AML, preferably containing HDAC (e.g., BEA (Gondo et al. 1997): BU/VP/Ara-C)	Either for use in AML
Cell source	Mobilized peripheral blood	Mobilized peripheral blood
Indication of CNS prophylaxis	ITT with MTX, hydrocortisone, and Ara-C, especially in those who presented relapse in CNS	ITT with MTX, hydrocortisone, and Ara-C, especially in those who presented relapse in CNS
Maintenance therapy post-HCT	Not proven, but conceivable that ATO + ATRA may be effective	Not proven, but conceivable that ATO + ATRA may be effective
Molecular monitoring	Recommended by RQ-PCR at least every 3 months for 2–3 years	Recommended by RQ-PCR at least every 3 months for 2–3 years

Table 70.3 Main series reported on HCT in APL

Reference	Study period – salvage therapy	No. of patients	Overall survival, %	Relapse rate, %
Sanz et al. (2007)	1993–2003 ATRA+CHT	Auto = 195 Alo = 137	Auto = 51[a] Alo = 59 P = ND	Auto = 37 Alo = 17 P = ND
De Botton et al. (2005)	1992–2001 ATRA+CHT	Auto = 50 Alo = 23	Auto = 60 Alo = 52 P = 0.04	Auto = 79[b] Alo = 92 P = 0.2
Kohno et al. (2008)	1999–2004 Miscelaneous	Auto = 15 Alo = 13	Auto = 76 Alo = 46 P = 0.2	Auto = 79[b] Alo = 92 NS
Lengfelder et al. (2015)	2003–2011 ATO-based	Auto = 60 Alo = 33	Auto = 77 Alo = 79 NS	Auto = 37 Alo = 39 NS
Pemmaraju et al. (2013)	1980–2010 Miscelaneous	Auto = 10 Alo = 17	Auto = 86 Alo = 49 NS	Auto = 30 Alo = 18 NS
Fujita et al. (2013)	1997–2002 ATRA+CHT	Auto = 6 Alo = 21	Auto = 83 Alo = 76 NS	Auto = 58[c] Alo = 10 P = 0.007
Alimoghaddam et al. (2011)	1989–2011 ATO-based	Auto = 11 Alo = 29	Auto = 47 Alo = 76 NS	NA
Holter Chakrabarty et al. (2014)	1995–2006 NA	Auto = 62 Alo = 232	Auto = 75 Alo = 54 P = 0.002	Auto = 30 Alo = 18 P = 0.4
Sanz et al. (2021)	2004–2018 NA	Auto = 341 Alo = 228	Auto = 82 Alo = 64 P = 0.001	Auto = 23 Alo = 28 P = 0.3

ATRA all trans-retinoic, *ATO* arsenic trioxide, *QT* quimioterapia, *NA* not available, *NS* no significant
[a] Leukemia-free survival
[b] Disease-free survival
[c] Crude relapse rate

Key Points

- The application of allo-HCT for patients with AML in first CR should be based on a risk-adapted strategy assessing both the risk of TRM and the risk of the AML.
- Allo-HCT remains a potentially curative treatment modality in fit patients with primary refractory AML providing a suitable donor can be identified rapidly and the predicted TRM is acceptable.
- Allo-HCT in acute promyelocytic leukemia is only indicated in specific cases (eg, ≥CR2 not in molecular remission or after previous auto-HCT).

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